

Sophia Robillard

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Education

University of Washington - Seattle, WA Sept 2023 - Jun 2026

B.Sc. in Earth and Space Sciences: Geosciences, with Departmental Honors

- 3.5 GPA
- Minor: Data Science
- Dean's List: Summer 2025, Spring 2025, Winter 2025, Summer 2024, Winter 2024

University of California, Santa Barbara - Santa Barbara, CA Sept 2022 - Jun 2023

Research Experience

Undergraduate Researcher - Δ^* IsoLab Sept 2024 - Present

Advised by Dr. Katharine Huntington - University of Washington

- Investigated $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$ isotopic variability and the petrography of lacustrine carbonates to determine biological and climatic influences on the stable isotope record and depositional patterns in the Colorado Plateau.
- Awarded *Best Undergraduate Poster Presentation* at the 2025 ESS Gala/Symposium; findings are contributing to a peer-reviewed manuscript currently in preparation.

Undergraduate Researcher - Isotope Geochemistry Laboratory Mar 2024 - Jun 2024

Advised by Dr. Bruce Nelson - University of Washington

- Learned techniques involving both purification of targeted elements through column chromatography in a clean sample preparation laboratory and high-precision isotopic analyses using MC-ICPMS.
- Collected preliminary $87\text{Sr}/86\text{Sr}$ isotope data from carbonate rock samples using mass-spectroscopy techniques.

Environmental Geochemistry Student - Geomicrobiology Laboratory Mar 2024 - Jun 2024

- Assessed natural waters for contamination and quality parameters (pH, dissolved oxygen, conductivity, temperature), alkalinity, iron (II and III), dissolved organic carbon, coliform bacteria.
- Performed alkalinity titrations, spectrophotometric analyses, and adsorption experiments relevant to low-temperature geochemistry.
- Gained experience in geochemical modeling softwares, PHREEQC and Equilibrium Diagrams, to model data found in environmental analyses. Reported findings, methods, and interpretations in cohesive laboratory reports.

Presentations and Publications

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- **Robillard, S.**, Heitmann, E.O., Germeau, H., Leite, M., Schauer, A., Huntington, K.W. (in prep). Formation Processes and Environments of Miocene Lacustrine Carbonates: Petrographic and Isotopic Implications for Paleoclimate Reconstruction on the Colorado Plateau, Seattle, WA, USA.
 - Heitmann, E.O., Huntington, K.W., **Robillard, S.**, Fischer, L., Jorgensen, A., Mat, S., Leite, M., Brigham, C., He, J., Douglass, J., Cassel, E., Gootee, B., Schauer, A. (in prep). Lacustrine carbonate isotope stratigraphy of the Miocene Bidahochi Formation records both short-term and long-term change, Navajo Nation, AZ, USA.
 - **Robillard, S.**, Heitmann, E.O., Germeau, H., Huntington, K.W., Schauer, A., Leite, M. (2025). Petrographic and Isotopic Constraints on the Processes and Environments of Miocene Lacustrine Carbonate Formation on the Colorado Plateau, Geological Society of America Connects 2025, National Conference, San Antonio, TX. (Poster)

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- **Robillard, S.**, Heitmann, E.O., Huntington, K.W., Schauer, A. (2025). Decoding Lacustrine Carbonates from the Colorado Plateau: How Depositional Environments and Processes Shape the Stable Isotope Record. University of Washington Earth & Space Sciences Gala, Seattle, WA. (Poster)

Honors and Awards

- Earth & Space Sciences Departmental Honors Program 2024 - Present
- Afton Woolley Crooks & James William Crooks Endowed Scholarship April 2025
in Geological Sciences - \$13,407
- “*Rising Rockstar*” Best Undergraduate Poster Presentation - \$200 April 2025

Service and Outreach

- Volunteer** - UW Rockin’ Out Outreach K-12 Jan 2025 - Present
- Design and lead engaging Earth and space science lessons, incorporating hands-on demonstrations to inspire curiosity and foster a deeper understanding of geoscience concepts.
 - Make complex topics accessible and exciting, encouraging students to ask questions, think critically, and explore their own scientific interests.
 - Create an inclusive and supportive learning environment, ensuring that students of all backgrounds feel empowered to engage with science and see themselves as future scientists.

- Field Assistant** - UW Department of Earth & Space Sciences Sept 2025
- Measured 50+ meters of stratigraphic sections in the Wenatchee Formation and created stratigraphic columns from detailed notes and drawings done in the field.
 - Collaborated with a small team to interpret the sedimentary rock record, collect samples for analysis, and record detailed outcrop observations.

- Teaching Assistant - Introduction to Geology (ESS 101)** Dr. Terry Swanson Jan 2025 - Mar 2025
- Volunteered in weekly lab sections, aiding 20+ students' comprehension of geologic concepts and providing feedback on lab assignments. Met weekly with teaching staff to discuss lesson plans, field trips, and criteria for grading lab exercises.

- After School Program Leader** - Isla Vista Elementary School K-6 Oct 2022 - Jun 2023
- Organized educational enrichment activities for 30+ elementary students, including founding a "Science Club" to conduct hands-on physics and chemistry experiments.

Professional Skills

Software Proficiency

- Python
- ArcGISPro
- PHREEQC
- HTML & CSS
- CloudCompare
- VSCode

Specialized Coursework

- Isotope Geochemistry
- Environmental Geochemistry
- Materials Science
- Advanced Calculus
- Mechanics/Electromagnetism
- Sedimentary Geology

Languages

- English (Native)
- French

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Professional Affiliations

- Student Member of the Geological Society of America (GSA)